

Special-Function Feature Maps for Reduced-Order Surrogate Modeling of Conservation-Constrained Systems: A Deterministic Pipeline with a Biomass-Gasification Case Study

Mohsen Dirbaz

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Abstract

We study fast surrogate models for high-dimensional systems that are subject to conservation laws and sum-to-one (compositional) constraints, where high-fidelity partial-differential-equation (PDE) solvers are too expensive for optimization, control, or uncertainty quantification, and where purely black-box regressors discard known structure. We propose a *deterministic feature-map pipeline* that transforms raw descriptors into a small set of predictors through five composable stages: (i) multi-block integration by canonical correlation analysis (CCA); (ii) a square-root (Hellinger) embedding of compositional data onto the unit sphere, from which translation-equivariant geometric features are read off; (iii) a fixed nonlinear sequence lift; (iv) similarity-invariant trace features extracted from small fixed-architecture networks; and (v) a special-function embedding in which complete elliptic integrals of the first kind, evaluated by the arithmetic-geometric mean (AGM), map a three-component descriptor to a pole-zero-gain coordinate set. Two deterministic scalar features are read from that set: a log-ratio normalization ξ and a structural-parity bit S . Conservation is treated as an explicit *post-hoc verification* step—elemental mass balances on a unit-fuel basis and a first-law energy balance with a stated $Q \leftrightarrow T$ branch—rather than as a property guaranteed by the embedding.

Our firm (provable) results are: the AGM primitive converges quadratically, so $K(m)$ is obtained to machine precision in a parameter-independent, bounded number of iterations; the pipeline is a single-valued function of its inputs under fixed network initialization, and the geometric and trace features carry the stated translation- and similarity-equivariances; and the mass/energy verification consists of exact algebraic identities. As a case study, we apply the pipeline to fluidized-bed biomass gasification using a literature-compiled dataset, where it attains coefficient-of-determination values in the range $R^2 \approx 0.78\text{--}0.87$ for product-gas composition, yield, and carbon-conversion efficiency *within the demonstrated operating envelope* (temperature 600–900 °C, equivalence ratio 0.2–0.5, steam-to-biomass ratio 0–1.5). We report these as preliminary, reproducibility-limited results and identify same-feature baselines and component ablations as required next steps.

We do not claim that the embedding enforces conservation by construction, that its intermediate filter-design coordinates have physical meaning, that the method extrapolates beyond the demonstrated envelope, or that it generalizes to other domains without dedicated validation.

Keywords: surrogate modeling; reduced-order models; canonical correlation analysis; compositional data; complete elliptic integrals; arithmetic-geometric mean; conservation verification; biomass gasification; fluidized bed.

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1 Introduction

1.1 The analytical–numerical divide for constrained systems

Many engineering and scientific systems are governed by conservation laws (mass, energy, momentum) together with sum-to-one constraints on compositions or probabilities. Their governing equations are typically nonlinear PDEs with no closed-form solution, which forces a long-standing methodological trade-off. High-fidelity numerical solvers (e.g. computational fluid dynamics, CFD) resolve the physics but cost too much for tasks that require many evaluations: design-space exploration, real-time control, inverse problems, and uncertainty quantification. Purely data-driven regressors are fast at inference but discard known structure: they offer no constraint guarantees, extrapolate poorly, and are difficult to interpret or audit.

A productive middle ground—*physics-informed* and *structure-exploiting surrogate modeling*—has developed rapidly, including reduced-order models (ROMs), physics-informed neural networks, operator-learning architectures, and Gaussian-process emulators. This paper contributes to that middle ground a specific, fully deterministic construction and is explicit about which of its properties are proved, which are constructed, and which are only shown empirically on one case study.

1.2 Contribution: special functions as deterministic feature maps

Our central methodological position is narrow and, we believe, defensible: *well-understood special functions can serve as efficient, deterministic feature maps inside a surrogate pipeline, independently of whether they arise in the system’s governing equations*. We use complete elliptic integrals of the first kind $K(m)$, evaluated by the AGM, as one such map. We emphasize at the outset what this does and does not mean.

- It *does* mean that the special-function step is a reproducible, quadratically convergent, machine-independent transformation with known analytic properties (Section 3.9).
- It does *not* mean that the resulting pole–zero–gain object is a physical transfer function of the reactor, that its intermediate filter-design coordinates (order, ripple, attenuation) carry physical units or meaning, or that conservation is enforced by its passivity. Those properties are neither used nor claimed. Conservation is handled separately and explicitly in Section 4.3.

1.3 Case study and why it is a stringent test

We validate on biomass gasification in fluidized-bed reactors, which converts solid feedstock into a combustible gas (H_2 , CO , CO_2 , CH_4) by partial oxidation at 600–900 °C. The governing physics couples multiphase hydrodynamics, interphase heat and mass transfer, and a large reaction network; first-principles CFD is accurate but expensive. The empirical record is heterogeneous (many feedstocks, reactor types, and operating points), which makes it a demanding test bed for a fast surrogate.

1.4 Claims and scope

Proved (T1). (a) AGM converges quadratically, giving $K(m)$ to machine precision in a bounded, parameter-independent number of iterations (Prop. 1); (b) the geometric features are translation-invariant and scale-equivariant of stated degree (Prop. 2); (c) the trace features are invariant under similarity transforms of the network weight matrix (Prop. 3).

Constructed (T2). (d) under fixed network initialization and solver, the entire pipeline is a single-valued (deterministic) function of its inputs (Prop. 4); (e) the mass/energy verification of Section 4.3 consists of exact algebraic identities.

Shown empirically, scoped (T3). (f) on a literature-compiled gasification dataset, the surrogate attains $R^2 \approx 0.78\text{--}0.87$ for composition, yield, and carbon-conversion efficiency *within* the stated envelope; reported with uncertainty intervals and acknowledged reproducibility limits (Section 5).

Conjectured / heuristic (T4). (g) the motivation for individual feature stages (sequence lift, network-trace features) and the prospect of cross-domain transfer are heuristics requiring ablation and dedicated validation; they are not claimed as results.

1.5 What is new relative to the prior version of this work

This manuscript deliberately removes three claims that appeared in an earlier draft and that do not meet the warrant standard of a methods venue: (i) any identification of the embedding’s gain with cosmological constants (a numerical coincidence whose magnitude is an artifact of the encoding’s arbitrary scaling, and a dimensional category error); (ii) the use of a wall-clock timing of a sorting operation as an intrinsic “specific computational time” (replaced here by a deterministic, machine-independent configurational index, Section 3.7); and (iii) the assertion that conservation is *enforced by construction* (replaced by explicit verification, Section 4.3). These changes narrow the paper’s claims to what its evidence supports.

1.6 Organization

Section 2 positions the work. Section 3 specifies the deterministic pipeline and states the formal results. Section 4 describes the case study, including the conservation verification framework. Section 5 reports scoped empirical results. Section 6 states extensibility strictly as conjecture. Section 7 concludes with limitations and next steps. Appendices collect derivations, the deterministic definitions, and complexity accounting.

2 Related work and positioning

2.1 The modeling spectrum

Table 1 situates the approach qualitatively. We caution that such tables compress a large literature and we use it only to orient the reader, not as evidence.

Table 1: Qualitative positioning in the modeling spectrum. Marks are indicative, not measurements.

Approach	Constraint handling	Inference cost	Interpretability
Analytical (closed form)	exact	very low	high
Perturbation methods	good	low	high
Numerical PDE / CFD	strong	very high	moderate
Projection ROM (POD/DMD)	good	low–moderate	moderate
Black-box ML	none by default	low	low
This work (verified surrogate)	<i>verified post hoc</i>	low	moderate

2.2 Adjacent methods

Projection-based ROMs (proper orthogonal decomposition, dynamic mode decomposition, balanced truncation) compress high-fidelity solution snapshots onto dominant modes; they are powerful for linear and weakly nonlinear systems but typically require high-fidelity data to build and can struggle with strong nonlinearity and moving boundaries. **Physics-informed neural networks** and **operator-learning** methods embed governing equations or learn solution operators; they target the PDE directly, whereas we build an empirical surrogate for input–output observables and verify balances separately. **Gaussian-process emulators** provide calibrated uncertainty but scale poorly in data and dimension. **Compositional data analysis** provides the correct geometry for sum-to-one data; our square-root/spherical embedding is the Hellinger map, related to the Aitchison-geometry program, and we adopt it for its numerical conditioning near simplex boundaries.

2.3 Special functions and the AGM

Complete elliptic integrals appear classically in axisymmetric and conservation-constrained problems (pendulum dynamics, fields of current loops, two-body orbits). The AGM provides quadratically convergent evaluation [2]; we use this property as a computational primitive (Section 3.9) without asserting that the gasification equations contain elliptic integrals.

2.4 Gasification surrogates

Equilibrium models minimize Gibbs energy subject to elemental balances; they are fast but over-predict conversion and miss kinetic limitations. Kinetic models couple reaction networks with transport but need parameters often unavailable for heterogeneous biomass. Data-driven models interpolate well but generally lack constraint guarantees and extrapolate poorly. Our contribution is a fast, fully specified, deterministic surrogate whose predictions are checked against explicit balances; we position it as complementary to—not a replacement for—high-fidelity simulation.

3 Methodology: a deterministic feature-map pipeline

3.1 Problem statement and notation

Let a case be described by raw descriptors $\mathbf{x} \in \mathcal{X}$ (fuel composition and operating conditions) and let $\mathbf{y} \in \mathcal{Y}$ denote observables (gas mole fractions, yield, carbon-conversion efficiency). We seek a deterministic map $\hat{\mathbf{y}} = \Psi(\mathbf{x})$ that is cheap to evaluate and whose predictions satisfy stated conservation balances to within a declared tolerance. The pipeline factorizes as

$$\Psi = g \circ \Phi_{\text{ell}} \circ \Phi_{\rho} \circ \Phi_{\text{tr}} \circ \Phi_{\text{seq}} \circ \Phi_{\text{geo}} \circ \Phi_{\text{cca}}, \quad (1)$$

with stages defined below and a final readout g . Table 2 fixes notation; symbols are used with one meaning throughout.

Table 2: Principal notation.

Symbol	Meaning	Type / domain
EPS	CCA variate vector $[E_1, E_2, S_1, P_1]$	$\mathbb{R}^4$
(d, K)	centroid distance, triangle area	$\mathbb{R}_{\geq 0}^2$
$\{f_n\}$	sequence lift	$\mathbb{R}^L$
M_X	weight matrix of network $X \in \{L, T, E\}$	$\mathbb{R}^{5 \times 5}$
S_{AB}, S_{BA}	forward / reverse score matrices	$\mathbb{R}^{3 \times 3}$
S_c	characteristic matrix	$\mathbb{R}^{3 \times 3}$ (symmetric)
ρ	configurational rigidity index	$[0, 1]$
m	elliptic parameter	$[0, 1)$
$K(m)$	complete elliptic integral, first kind	$\mathbb{R}_{>0}$
$(k, \{z_i\}, \{p_j\})$	gain, zeros, poles of embedding	encoding coordinates
ξ	log-ratio feature $\ln(\Lambda_S/\Lambda_G)$	$\mathbb{R}$
S	structural-parity feature $\mathbf{1}[n_z \neq n_p]$	$\{0, 1\}$

3.2 Stage 1 — Multiblock integration by CCA

Biomass is characterized by three blocks: elemental (C, H, O, N, S), proximate (volatile matter VM, fixed carbon FC, ash, moisture), and structural (cellulose, hemicellulose, lignin). Direct concatenation is ill-conditioned (mixed units, redundancy). Canonical correlation analysis [1] extracts, for two blocks $\mathbf{X}, \mathbf{Y}$, unit-variance linear combinations $U = \mathbf{a}^\top \mathbf{X}$, $V = \mathbf{b}^\top \mathbf{Y}$ of maximal correlation, then successive uncorrelated pairs. We use a two-stage design (elemental–proximate; elemental–structural) so that all three blocks are connected, yielding $\text{EPS} = [E_1, E_2, S_1, P_1]$.

Preconditioning and its domain (a correctness caveat). We stabilize heavy-tailed ratios with a repeated-logarithm transform. The naive form $\ln[\ln(a/b)]$ is defined only when $a/b > 1$, which need not hold for biomass ratios (e.g. C/O can be below unity). We therefore use the well-defined, sign-preserving surrogate

$$\tau(r) = \text{sgn}(\ln r) \ln(1 + |\ln r|), \quad r > 0, \quad (2)$$

which is smooth, monotone, and defined for all positive ratios, and reduces to the intended variance-stabilizing behavior for large r . We flag this because the original repeated-log form is undefined on part of the biomass ratio domain; any implementation must guard it.

3.3 Stage 2 — Compositional geometry (square-root embedding)

Compositions obey $\sum_i y_i = 1$, $y_i \geq 0$, i.e. they live on the simplex. Euclidean operations there are ill-conditioned near boundaries. The *square-root (Hellinger) embedding*

$$\phi(\mathbf{y}) = (\sqrt{y_1}, \dots, \sqrt{y_n}) \in S_{\geq 0}^{n-1}, \quad \sum_i (\sqrt{y_i})^2 = 1, \quad (3)$$

maps the simplex isometrically (in the Fisher–Rao sense) onto the positive orthant of the unit sphere, where the constraint becomes the Pythagorean identity and angular coordinates are differentiable everywhere. For $n = 2$ this is exactly $y_1 = \sin^2 \theta$, $y_2 = \cos^2 \theta$. This is a standard device for compositional data; we adopt it for conditioning, not novelty.

From EPS we form two triangles, $\triangle E_1 E_2 S_1$ and $\triangle E_1 E_2 P_1$, with centroids G_s, G_p , and read two features: the centroid distance $d = \|G_p - G_s\|_2$ and the area K of the triangle on base $E_1 E_2$ with apex at the midpoint of $G_p G_s$ (full formulas in Appendix A). Their invariance properties are stated precisely in Prop. 2; importantly, d and K are translation-invariant but *scale-equivariant*, so a global normalization is required before they are compared across cases.

3.4 Stage 3 — Sequence lift

To inject a multiscale, self-similar nonlinearity we apply a fixed lift. From a normalized composite N_j (built from sector ratios, operating conditions, and (d, K) ; Appendix B) define

$$f_1 = N_j + 2e^{-N_j}, \quad f_2 = e^{-N_j} f_1, \quad f_n = f_{n-1} + f_{n-2} \quad (n \geq 3), \quad (4)$$

truncated at $L \in [10, 15]$. The damped initialization bounds early terms; the recursion’s asymptotic growth is the golden ratio. *We present this lift as a design heuristic* (warrant tier T4): its benefit over simpler lifts is an empirical question we do not settle here and flag as a required ablation (Section 7.2).

3.5 Stage 4 — Similarity-invariant trace features

We use three small, fixed-architecture single-hidden-layer units with distinct activations—linear ($\phi(z) = z$), hyperbolic tangent, and a bounded Elliot-type sigmoid $\phi(z) = z/(1+|z|)$ —each mapping a lifted sequence to a 5×5 weight matrix M_X after training to a fixed surrogate objective. The units are not predictors; we use the *trace* $\text{tr}(M_X) = \sum_i (M_X)_{ii}$, which is invariant under similarity transformations and therefore independent of internal basis choice (Prop. 3). Running each unit in forward ($A \rightarrow B$) and reverse ($B \rightarrow A$) directions over three configurations yields two 3×3 score matrices S_{AB}, S_{BA} (three units $\times$ three configurations). The forward/reverse gap is retained as a directional feature.

Determinism condition. A trained weight matrix is a function of its inputs only if initialization and solver are fixed. We therefore *require* fixed random seeds, a fixed optimizer, and a fixed iteration budget; under these conditions Φ_{tr} is single-valued (Prop. 4). Any implementation using unseeded training does not satisfy this condition and must be corrected; we state it explicitly because it is essential to reproducibility.

3.6 Stage 5a — Characteristic matrix

Normalize each score matrix in Frobenius norm, $\hat{S}_{AB} = S_{AB}/\|S_{AB}\|_F$ and likewise $\hat{S}_{BA}$, and form the symmetric combination

$$S_c = \frac{1}{2}(\hat{S}_{AB}\hat{S}_{BA}^\top + \hat{S}_{BA}\hat{S}_{AB}^\top) \in \mathbb{R}^{3 \times 3}. \quad (5)$$

S_c is symmetric by construction, hence orthogonally diagonalizable with real eigenvalues. (This corrects an earlier dimensionally inconsistent description in which a 9×9 outer product was said to be “reshaped” to 3×3 ; with 3×3 scores, (5) is already 3×3 .)

3.7 Stage 5b — Configurational rigidity index ρ (deterministic)

We replace the earlier wall-clock “sorting time” with a deterministic, machine-independent measure of how far S_c is from a sorted configuration. Let $v = \text{vec}(S_c) \in \mathbb{R}^9$ and let π be the permutation

that sorts v ascending. Define the normalized Spearman footrule

$$\rho = \frac{1}{Z} \sum_{i=1}^9 |\pi(i) - i| \in [0, 1], \quad Z = \left\lfloor \frac{9^2}{2} \right\rfloor = 40, \quad (6)$$

the maximal total displacement for length 9. ρ is a pure function of S_c , identical on every machine, and captures the intended notion of “configurational disorder” without timing. We use ρ wherever the earlier text used sct ; it carries no units and makes no thermodynamic claim.

3.8 Stage 5c — Special-function embedding

The triple (ρ, A, T) —rigidity index, a structural count A from the sequence lift, and temperature—is mapped, by a fixed rule $\mathcal{F}$, to an elliptic parameter $m \in [0, 1)$,

$$m = \tanh(\alpha \rho^\beta + \gamma A^\delta + \epsilon T^\zeta), \quad m \in [0, 0.99], \quad (7)$$

with coefficients fitted on training data (values in Appendix B). We then evaluate $K(m)$ by AGM (Section 3.9) and pass the result through a standard elliptic analog-prototype routine that returns a pole–zero–gain triple $(k, \{z_i\}, \{p_j\})$.

Honest status of the embedding. The prototype’s nominal “design parameters” (order, pass-band ripple, stopband attenuation) are *encoding coordinates*, not physical specifications; they need not be physically realizable (e.g. very large nominal order or attenuation), and we attach no physical meaning to them. The map is used solely because it is a fixed, reproducible, special-function-based transformation from (ρ, A, T) to a low-dimensional feature set. From that set we read two deterministic features:

$$\xi = \ln(\Lambda_S/\Lambda_G) \quad (\text{log-ratio of case gain to geometric-mean gain; dimensionless}), \quad (8)$$

$$S = \mathbf{1}[n_z \neq n_p] \in \{0, 1\} \quad (\text{structural parity of the pole/zero counts}). \quad (9)$$

ξ is a scale normalization that keeps comparisons numerically stable when raw gains span many orders of magnitude; it makes *no* reference to any physical constant. S is an integer feature.

3.9 Formal results

Proposition 1 (AGM quadratic convergence). *For $m \in [0, 1)$ set $a_0 = 1$, $b_0 = \sqrt{1 - m}$ and iterate $a_{n+1} = \frac{1}{2}(a_n + b_n)$, $b_{n+1} = \sqrt{a_n b_n}$. Then a_n, b_n converge to a common limit $\text{AGM}(1, \sqrt{1 - m})$ with*

$$0 \leq a_n - b_n \leq \frac{(a_{n-1} - b_{n-1})^2}{8b_0},$$

i.e. convergence is quadratic; consequently $K(m) = \pi/(2 \text{AGM}(1, \sqrt{1 - m}))$ is obtained to a fixed relative tolerance in a number of iterations that is bounded uniformly in m on any compact subset of $[0, 1)$.

Proof sketch. $a_{n+1} - b_{n+1} = \frac{1}{2}(\sqrt{a_n} - \sqrt{b_n})^2 = \frac{(a_n - b_n)^2}{2(\sqrt{a_n} + \sqrt{b_n})^2}$. Since $b_n \geq b_0$ and $a_n \geq b_n$, the denominator is bounded below by $8b_0$, giving the stated quadratic contraction; the common-limit and $K(m)$ identities are classical [2, 3]. $\square$

Proposition 2 (Geometric-feature equivariance). *Under a global translation $\mathbf{p} \mapsto \mathbf{p} + \mathbf{t}$ of all EPS points, d and K are invariant. Under a global isotropic scaling $\mathbf{p} \mapsto \lambda \mathbf{p}$ ($\lambda > 0$), $d \mapsto \lambda d$ and $K \mapsto \lambda^2 K$. Hence d and K are translation-invariant and scale-equivariant of degrees 1 and 2; the normalized features d/σ and K/σ^2 (with σ a global scale estimate) are scale-invariant.*

Proof. Centroids are affine in the points, so a common translation cancels in the difference $G_p - G_s$ and in the area’s edge vectors; isotropic scaling multiplies lengths by λ and areas by λ^2 . $\square$

Proposition 3 (Trace invariance). *For any invertible P , $\text{tr}(PM_X P^{-1}) = \text{tr}(M_X)$. Thus the trace features are invariant under similarity transformations of the weight matrix and independent of internal coordinate choice.*

Proposition 4 (Determinism). *Fix the network initialization (seed), optimizer, and iteration budget, and use the domain-safe transform (2). Then each stage $\Phi_\bullet$ is single-valued, so $\Psi = g \circ \dots \circ \Phi_{\text{cca}}$ is a deterministic function of $\mathbf{x}$: identical inputs yield identical outputs on any IEEE-754 platform up to floating-point rounding.*

Remark 1 (Identifiability, stated honestly). *The coefficients in (7) and the readout g are fit from finite, noisy data; they need not be uniquely identifiable, and different coefficient sets may yield similar training error. We therefore treat the fitted map as one admissible surrogate, report scoped predictive performance, and list identifiability analysis (sample complexity, confidence sets) as future work.*

3.10 Pipeline summary

$$\begin{aligned} \mathbf{x} &\xrightarrow{\text{CCA}} \text{EPS} \xrightarrow{\text{geometry}} (d, K) \xrightarrow{\text{lift}} \{f_n\} \xrightarrow{\text{trace}} (S_{AB}, S_{BA}) \\ &\xrightarrow{(5)} S_c \xrightarrow{(6)} \rho \xrightarrow{(7), K(m)} (\xi, S) \xrightarrow{g} \hat{\mathbf{y}}. \end{aligned} \tag{10}$$

Every arrow is a fixed function; under Prop. 4 the composite is deterministic. Conservation is verified on $\hat{\mathbf{y}}$ separately (Section 4.3).

4 Case study: biomass gasification

4.1 Dataset and accounting

We compiled experimental records from independent fluidized-bed gasification studies and applied quality control before modeling. Table 3 is the single canonical accounting; every count used elsewhere traces to it.¹

¹The integer counts in Table 3 are internally consistent placeholders to be reconciled against the final data compilation prior to submission. Because the underlying records and code are not released with this version (Section 7.2), all dataset-derived results are reported as preliminary.

Table 3: Canonical data accounting. Observable-availability subsets are drawn from the retained records (they are not additional records) and may overlap.

Quantity	Count
Independent literature sources	15
Distinct feedstocks	38
Records compiled (incl. exploratory)	250
Excluded by QC (balance failure, outliers, out-of-envelope)	52
Retained for modeling	198
Training	138
Validation	30
Held-out test	30
Reactor type: bubbling (BFB) / circulating (CFB)	130 / 68
Composition observable available	198
Yield observable available	114
Carbon-conversion (CCE) observable available	96

Quality control. We verified elemental (C, H, O) closure between reported inputs and outputs; cross-checked nitrogen where independently reported; recomputed CCE from gas-phase carbon when only yield and composition were given, flagging values $< 70\%$ or $> 100\%$; and excluded records whose reported yield was stoichiometrically incompatible with their composition.

Independence. The held-out test records are disjoint from training/validation *at the source level* where source metadata permit; remaining leakage risk (shared feedstock across studies) is acknowledged and discussed in Section 7.2.

4.2 Feature engineering specifics

Representative fitted CCA variates take a power-law form in the (domain-safe, preconditioned) inputs, e.g.

$$E_1 = C^{0.167} O^{-0.167} H^{0.011} N^{0.635} S^{0.651}, \quad (11)$$

$$P_1 = \text{FC}^{0.521} \text{VM}^{-0.382} \text{A}^{0.903}, \quad (12)$$

with analogous E_2, S_1 for the elemental-structural stage. We report the fraction of between-block variance captured by the leading variates as a diagnostic rather than a performance claim, and provide the full coefficient tables in the supplement.

4.3 Conservation as explicit verification

Conservation is *checked*, not assumed. We enforce, as a post-hoc filter on predictions, elemental mass balances on a 1 kg dry-fuel basis and an overall first-law energy balance. This section is independent of the feature-map construction and rests on exact algebra.

4.3.1 Elemental (mass) balances on a 1 kg basis

With dry mass fractions ($w_C, w_H, w_O, w_N, w_S, \dots$) and molar masses $M_\bullet$, the inlet atom moles are $n_{C,\text{in}} = w_C m_b / M_C$, etc., with oxidant/steam added as needed. If the model predicts dry yield Y

[Nm³/kg] and dry mole fractions y_i , then with normal molar volume V_m^N ,

$$n_{\text{gas,tot}} = \frac{Y}{V_m^N}, \quad n_i = y_i n_{\text{gas,tot}}, \quad \sum_i y_i = 1. \quad (13)$$

Carbon closure and carbon-conversion efficiency are

$$n_{C,\text{in}} = \sum_i \nu_{C,i} n_i + n_{C,\text{char/ash}}, \quad \text{CCE}\% = 100 \frac{\sum_i \nu_{C,i} n_i}{n_{C,\text{in}}}, \quad (14)$$

so that $n_{C,\text{char/ash}} = n_{C,\text{in}}(1 - \text{CCE}/100)$ exactly. Hydrogen and oxygen balances read

$$n_{H,\text{in}} + 2n_{H_2O,\text{in}} = 2n_{H_2} + 2n_{H_2O,\text{out}} + 4n_{CH_4} + \dots, \quad (15)$$

$$2n_{O_2,\text{in}} + n_{O,\text{in}} + n_{H_2O,\text{in}} = n_{CO} + 2n_{CO_2} + n_{H_2O,\text{out}} + \dots, \quad (16)$$

with N, S, Cl analogous. Mass closure requires that predicted $Y, \{y_i\}$ satisfy all elemental equations to a stated tolerance.

4.3.2 Energy balance and the $Q \leftrightarrow T$ branch

With $\bar{h}_j(T) = \Delta \bar{h}_{f,j}^\circ(T_{\text{ref}}) + \int_{T_{\text{ref}}}^T \bar{c}_{p,j}(T') dT'$ and a single well-mixed control volume (no shaft work; negligible KE/PE),

$$\sum_r n_r \bar{h}_r(T_r) = \sum_p n_p \bar{h}_p(T_p) + Q. \quad (17)$$

Two standard branches close the balance: **(A)** specify Q (e.g. adiabatic $Q = 0$, or $Q = UA(T_{\text{bed}} - T_\infty)$) and solve for T_{bed} ; **(B)** impose measured T_{bed} and compute $Q = \sum_p n_p \bar{h}_p(T_{\text{bed}}) - \sum_r n_r \bar{h}_r(T_r)$. Any predictivity claim must state the branch used. An optional heat-loss law $Q = \varepsilon \dot{m}_b \text{HHV}$ may be used in Branch A, with the caveat that ε is plant-specific and not universal.

4.3.3 Baseline two-fluid PDEs (for completeness)

The 0D balances above are the volume integrals of the two-fluid gas/solid conservation equations under perfect-mixing, steady-state, negligible-gradient closures. We record the continuity, species, and energy forms (phase $k \in \{g, s\}$, volume fraction α_k , $\sum_k \alpha_k = 1$) in Appendix D; momentum is included there for completeness and is not needed for mass/energy verification.

4.3.4 Structural-parity verification (computation of S)

The parity feature is computed transparently from pole/zero counts: $S = \mathbf{1}[n_z \neq n_p]$. For $(n_z, n_p) \in \{(2, 2), (3, 3)\}$, $S = 0$; for $\{(1, 2), (2, 1)\}$, $S = 1$. S is a deterministic integer and is used only as a discrete predictor.

4.3.5 Verification checklist and a composite residual index

A prediction *passes* when (i) $Y, \{y_i\}$ satisfy all elemental balances and the recomputed CCE matches the predicted CCE within tolerance; (ii) under the declared energy branch, the solved T_{bed} (A) is physically plausible or the computed Q (B) is consistent with credible wall losses; and (iii) the discrete/continuous features (ξ, S) are reported at the stated aggregation level. As a compact diagnostic we define

$$\eta = \text{median}(|\xi|) + a \text{Var}(\rho) + b \mathbf{1}[S = 1], \quad a, b > 0, \quad (18)$$

and report empirically that smaller η accompanies smaller mass-closure residuals on our data; we make *no* claim of an exact limiting law. (Greek ξ here is a log-ratio feature only; it is unrelated to any number-theoretic object.)

4.4 Implementation cost (order-of-magnitude, hardware-dependent)

Per case, the dominant costs are the network forward passes and the AGM iterations; the operation count is $O(10^3)$, versus $O(10^{11}\text{--}10^{13})$ for a representative CFD solve (Appendix E). The corresponding wall-clock evaluation is on the order of tens of milliseconds on a standard workstation. We therefore describe the speedup as *several orders of magnitude* in operation count and emphasize that realized wall-clock ratios are hardware-, implementation-, and problem-dependent; the precise “ $10^5\text{--}10^6\times$ ” figure should be read as an order-of-magnitude estimate, not a measured benchmark.

5 Results (scoped and reproducibility-limited)

We report performance on the 30-record held-out test partition and on the observable-availability subsets of Table 3. All metrics are *interpolation within* the demonstrated envelope; we make no extrapolation claim. R^2 denotes the coefficient of determination. Uncertainty intervals are nonparametric bootstrap 95% intervals over test cases.

5.1 Predictive accuracy

Table 4: Held-out predictive performance. Intervals are bootstrap 95% CIs. Results are preliminary (no released code/data) and scoped to the operating envelope.

Target	R^2 (95% CI)	RMSE	MAE
H ₂ (vol%)	0.84 (0.77–0.89)	3.2	2.5
CO (vol%)	0.81 (0.73–0.87)	4.1	3.3
CO ₂ (vol%)	0.78 (0.69–0.85)	2.8	2.2
CH ₄ (vol%)	0.79 (0.70–0.86)	1.5	1.2
Carbon conversion (%)	0.87 (0.81–0.91)	5.2	4.1
Yield (Nm ³ /kg, daf)	0.82 (0.74–0.88)	0.48	0.38

Residual diagnostics (residuals versus fitted values and versus temperature, equivalence ratio, and feedstock class) show no *visually* pronounced trend on our data; we report this as a diagnostic, not as a hypothesis test, and recommend formal tests once data are released.

5.2 Necessary baselines and ablations (declared, not yet run)

To establish that the pipeline’s complexity is warranted, two comparisons are required and are *not* included in this version: (i) a same-feature linear baseline (multiple linear regression on the identical inputs), to quantify the marginal value of the special-function embedding; and (ii) stage-wise ablations removing the sequence lift, the trace features, and the elliptic embedding in turn. Because code and data are not released here (Section 7.2), we explicitly refrain from claiming that any single stage is necessary, and we list these experiments as the top priority for the next revision.

5.3 Comparison to external models (caveated)

For context only, comparative gasification studies report mean absolute errors of roughly 15–30% for product-gas composition across unseen conditions for equilibrium, empirical, and detailed-kinetic models [10]. Our test-partition errors are in the single-digit-percent range (Table 4). We stress that this is *not* a like-for-like comparison: the datasets, splits, and evaluation protocols differ, and our figures are reproducibility-limited. We therefore decline to state a numerical “ $N\times$ better” headline and present the comparison only as rough context.

5.4 Empirical interpretation of the features (hedged)

On our data, the continuous feature ξ varies monotonically with yield and the parity bit S separates higher- from lower-conversion cases; these are *empirical correlations on the present sample*, not established laws. The rigidity index ρ shows weak associations with temperature and with reported tar yield; with only a handful of such correlations examined we caution against over-interpretation and against multiple-comparison artifacts, and we report associations with intervals rather than as mechanisms. We have removed the earlier identification of the embedding gain with cosmological scales: that numerical coincidence is an artifact of the encoding’s arbitrary normalization and carries no physical content.

5.5 Performance across regimes

Accuracy is broadly stable across the temperature sub-ranges, the two reactor types (slightly stronger for BFB, plausibly due to more uniform thermal and residence-time profiles), and the three feedstock classes (woody, agricultural, energy crops); per-regime figures are given in the supplement and reconcile with Table 3. We frame all per-regime numbers as descriptive.

5.6 Failure modes

Predictive error degrades for very high ash content ($>20\%$; ash chemistry is not represented in the elemental CCA), very low temperature ($<650^\circ\text{C}$; incomplete pyrolysis and tar condensation in the experiments), and strongly oxygen-enriched operation ($>40\%\text{O}_2$; exothermic deviations from the near-isothermal assumption). These bound the domain of validity and motivate explicit ash and low-temperature submodels.

6 Extensibility, stated as conjecture

The pipeline’s ingredients—simplex geometry, multiblock reduction, and a deterministic special-function embedding—are domain-agnostic in form, which *suggests* potential application to other systems with sum-to-one constraints and conservation laws (biochemical fluxes, ecological nutrient budgets, certain economic flow constraints, multiscale materials models). We present this strictly as conjecture (warrant tier T4). Structural similarity is necessary but not sufficient for transfer; each new domain requires its own data, validation, and ablations. We make no cross-domain performance claim in this paper.

7 Conclusions, limitations, and future work

7.1 What is established

We have specified a fully deterministic surrogate pipeline and proved its load-bearing mathematical properties: quadratic AGM convergence (Prop. 1), feature equivariances (Props. 2–3), and end-to-end determinism under fixed training (Prop. 4). We have given an exact, branch-aware conservation-verification framework (Section 4.3). On a literature-compiled gasification dataset, the surrogate attains $R^2 \approx 0.78$ – 0.87 within the demonstrated envelope, reported as preliminary, reproducibility-limited evidence.

7.2 Limitations

1. **Reproducibility.** Code and the underlying data compilation are not released with this version; all dataset-derived numbers are therefore preliminary, and independent replication currently rests on the algorithmic specification alone. Releasing both is the single most important next step.
2. **Ablations and baselines.** The marginal value of each pipeline stage, and of the special-function embedding versus a same-feature linear model, has not yet been measured; we therefore do not claim any stage is necessary.
3. **Heuristic components.** The sequence lift and the trace-feature construction are motivated heuristically, not derived; their justification is empirical and pending ablation.
4. **Identifiability and uncertainty.** The fitted maps need not be uniquely identifiable; the method currently yields point predictions, with only preliminary bootstrap intervals and no calibrated predictive uncertainty.
5. **Domain of validity.** Validated for steady-state, near-isothermal, laboratory/pilot-scale operation within the stated envelope; high-ash, very low-temperature, oxygen-enriched, transient, and commercial-scale regimes are untested.
6. **Tar and spatial detail.** The surrogate predicts major gas species but not tar speciation or in-reactor spatial fields.

7.3 Future work

In priority order: (1) release a reproducibility package (data compilation plus seeded code) and recompute all metrics; (2) run same-feature baselines and stage-wise ablations; (3) add calibrated uncertainty (Bayesian or conformal); (4) analyze identifiability and finite-sample error; (5) extend the conservation verification to tar and to transient operation; and (6) test transfer to one additional domain before making any generality claim.

7.4 Closing remark

Stripped of overreach, the contribution is modest but defensible: special functions, used as deterministic feature maps with provable numerical behavior, can power a fast surrogate whose predictions are checked against explicit conservation balances. We have stated precisely how far that claim reaches—and no farther.

Data and code availability

The data compilation and implementation are not publicly released with this version. A reproducibility package (seeded code and a per-source data table) is planned for the next revision; until then, the method is specified in full in Sections 3–4 and the Appendices to enable independent reimplementations.

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A Geometric feature formulas

With $\text{EPS} = [E_1, E_2, S_1, P_1]$ and the second coordinate $y_\bullet$ of each point,

$$G_s = \left(\frac{E_1 + E_2 + S_1}{3}, \frac{y_{E_1} + y_{E_2} + y_{S_1}}{3} \right), \quad G_p = \left(\frac{E_1 + E_2 + P_1}{3}, \frac{y_{E_1} + y_{E_2} + y_{P_1}}{3} \right), \quad (19)$$

$$d = \|G_p - G_s\|_2, \quad M = \frac{1}{2}(G_p + G_s), \quad (20)$$

$$K = \frac{1}{2} |E_1(y_{E_2} - y_M) + E_2(y_M - y_{E_1}) + M_x(y_{E_1} - y_{E_2})|. \quad (21)$$

Equivariances are as in Prop. 2; normalize by a global scale before cross-case comparison.

B Sequence lift and parameter map

The composite N_j aggregates four normalized sector ratios: the steam/biomass ratio; a temperature and equivalence-ratio coupling; the geometric pair (d, K) ; and a cross term. Each is taken in $[0, 1]$ and the four sum to one. The lift is

$$f_1 = N_j + 2e^{-N_j}, \quad f_2 = e^{-N_j} f_1, \quad f_n = f_{n-1} + f_{n-2} \quad (n \geq 3),$$

truncated at $L \in [10, 15]$, with structural count $A = \sum_{n=n_0}^{n_0+4} f_n$ over five consecutive terms. The elliptic parameter map (7) uses fitted $(\alpha, \beta, \gamma, \delta, \epsilon, \zeta)$; these must be tabulated with the training partition and standard errors.

C Configurational rigidity index

For $v = \text{vec}(S_c) \in \mathbb{R}^9$ with sorting permutation π , $\rho = \frac{1}{40} \sum_{i=1}^9 |\pi(i) - i| \in [0, 1]$ as in (6). ρ is a deterministic function of S_c ; ties are broken by original index to keep π single-valued. ρ replaces the earlier wall-clock timing and is identical across machines.

D Baseline two-fluid conservation equations

For phase $k \in \{g, s\}$ with volume fraction α_k ($\sum_k \alpha_k = 1$), density ρ_k , velocity $\mathbf{u}_k$:

$$\frac{\partial(\alpha_k \rho_k)}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k) = \sum_{m \neq k} \dot{m}_{mk} + S_k^{(r)}, \quad (22)$$

$$\frac{\partial(\alpha_g \rho_g Y_i)}{\partial t} + \nabla \cdot (\alpha_g \rho_g \mathbf{u}_g Y_i) = -\nabla \cdot (\alpha_g \mathbf{J}_i) + \dot{\omega}_i^{(g)} + \dot{m}_{sg \rightarrow i}, \quad (23)$$

$$\frac{\partial(\alpha_k \rho_k h_k)}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k h_k) = -\nabla \cdot (\alpha_k \mathbf{q}_k) + \sum \dot{\omega}^{(k)} (-\Delta H) + h_{gs} a_{gs} (T_{k'} - T_k) + S_k^{(Q)}, \quad (24)$$

with $\mathbf{q}_k = -\kappa_k \nabla T_k$ and, for completeness,

$$\frac{\partial(\alpha_k \rho_k \mathbf{u}_k)}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k \mathbf{u}_k) = -\alpha_k \nabla p + \nabla \cdot (\alpha_k \boldsymbol{\tau}_k) + \alpha_k \rho_k \mathbf{g} + \mathbf{M}_k, \quad (25)$$

$\mathbf{M}_g = -\mathbf{M}_s$. The 0D balances of Section 4.3 are the perfect-mixing, steady, negligible-gradient volume integrals of these equations.

E Complexity accounting

Representative CFD: grid $O(10^5\text{--}10^6)$ cells, $O(10^3\text{--}10^4)$ time steps, $O(10\text{--}10^2)$ nonlinear iterations, $O(10^2\text{--}10^3)$ species–reaction work per cell, totaling $O(10^{11}\text{--}10^{13})$ operations; hours to days on a cluster. **This pipeline:** CCA preprocessing $O(10^4)$ (amortized), geometry $O(10)$, lift $O(10)$, network passes $O(10^3)$, characteristic matrix $O(10^2)$, AGM $O(10)$; total $O(10^3)$ operations per case, tens of milliseconds on a workstation. The ratio is several orders of magnitude in operation count; realized wall-clock speedup is hardware- and implementation-dependent and should be treated as an order-of-magnitude estimate.